SRINIVAS UNIVERSITY

INSTITUTE OF ENGINEERING & TECHNOLOGY

MUKKA, MANGALURU-574146

MINI PROJECT REPORT

ON

Skill Verse AI

Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN

COMPUTER SCIENCE & ENGINEERING

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Department of Cyber Security & Cyber Forensics

CERTIFICATE

This is to certify that the Technical Seminar entitled ***"skill_verse AI"*** is a bonafide work carried out by ***P D Prajwal, Pratham M Naik*** bearing the **USN** ***01SU23CS126, 01SU23CS137*** in the partial fulfillment for the award of **Bachelor of Technology** in **Computer Science & Engineering** of the **Srinivas University-Institute of Engineering & Technology** during the year **2025-26.** It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The mini project report has been approved as it meets the academic criteria outlined for the said degree.

Name & Signature	Name & Signature
Mr. Yogesh Bandewar **(Project Coordinator)**	**Dr.Shifana Begaum** **(HOD, CSE)**

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DECLARATION

We , ***P D Prajwal, Pratham M Naik*** the student of 6th semester, **Bachelor of Technology** in **Computer Science & Engineering** of the **Srinivas University-Institute of Engineering & Technology**, Mukka, hereby declare that the mini project report entitled ***"Skill_Verse AI"*** has been successfully completed by us in partial fulfillment of the requirements for the award of degree in **Bachelor of Technology** in **Computer Science & Engineering of Srinivas University- Institute of Engineering & Technology** and no part of it has been submitted for the award of degree or diploma in any university or institution previously.

Date: 25-03-2026
Place: Mukka

ABSTRACT

SkillVerse AI is an AI-powered platform designed to improve job readiness by analyzing resumes and providing actionable career guidance. The system utilizes the Gemini API to evaluate user-uploaded resumes based on Applicant Tracking System (ATS) standards, generating a comprehensive report that includes an ATS score, keyword analysis, formatting evaluation, and role-specific feedback.

The platform also supports job-specific optimization by allowing users to upload a job description along with their resume. It performs a comparative analysis to identify missing skills, keyword gaps, and content misalignment, and suggests targeted improvements to enhance the chances of selection.

Additionally, SkillVerse AI provides a personalized roadmap that guides users on skill development, project building, and profile enhancement based on their career goals. By combining intelligent analysis with structured recommendations, the system helps users transform their resumes into optimized, job-focused profiles and systematically improve their employability.

ACKNOWLEDGEMENT

It is our pleasure to acknowledge for all those who have provided guidance, inspiration and encouragement for the successful completion of my mini project.

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Chapter-1

INTRODUCTION

In today's competitive job market, securing employment requires more than just qualifications; it demands a well-structured resume that aligns with Applicant Tracking System (ATS) requirements and specific job roles. Most companies rely on ATS software to filter resumes before human review, causing many qualified candidates to be rejected due to poor keyword optimization, formatting issues, or lack of alignment with job descriptions. As a result, job seekers often struggle to understand why their applications fail and how to improve them effectively.

SkillVerse AI is developed to address this gap by providing an intelligent, automated solution for resume evaluation and enhancement. The platform uses the Gemini API to analyze resumes and generate detailed insights, including ATS compatibility scores, keyword relevance, and structural feedback. Unlike traditional resume checkers, SkillVerse AI goes a step further by allowing users to upload job descriptions and performing a comparative analysis to identify gaps between candidate profiles and employer expectations.

Furthermore, the system offers a personalized roadmap that guides users in improving their skills, projects, and overall profile based on career goals and industry demands. This transforms the resume from a static document into a dynamic tool for career growth. By integrating artificial intelligence with practical recommendations, SkillVerse AI aims to simplify the job preparation process and increase the chances of candidate success in modern recruitment systems.

Chapter 2

PROBLEM STATEMENT

In modern recruitment systems, a significant number of resumes are filtered out by Applicant Tracking Systems (ATS) before reaching human recruiters. Many candidates, especially students and early-career professionals, lack awareness of ATS requirements such as keyword optimization, proper formatting, and alignment with job descriptions. As a result, even qualified individuals face rejection due to poorly structured or non-optimized resumes.

Existing resume evaluation tools are either too generic, lack personalized feedback, or fail to provide actionable insights tailored to specific job roles. Additionally, most platforms do not effectively bridge the gap between a candidate's current profile and the expectations outlined in job descriptions, leaving users without clear guidance on how to improve.

There is a need for an intelligent system that can analyze resumes, evaluate their compatibility with ATS standards, compare them with job descriptions, and provide precise, actionable recommendations along with a structured roadmap for improvement. Such a solution should help users systematically enhance their resumes and increase their chances of success in competitive hiring processes

Chapter 3

OBJECTIVES OF THE PROJECT

1. To develop an AI-based resume analysis system
   Design and implement a platform that uses the Gemini API to evaluate resumes based on ATS standards, including keyword relevance, formatting, and content quality.

2. To generate an ATS compatibility score with detailed feedback
   Provide a structured report that includes a numerical ATS score along with section-wise analysis such as skills, experience, and keyword optimization.

3. To perform job description and resume matching
   Enable users to upload job descriptions and compare them with their resumes to identify missing skills, keyword gaps, and alignment issues.

4. To recommend targeted resume improvements
   Suggest precise and actionable changes such as adding relevant keywords, modifying content, and restructuring sections to improve selection chances.

5. To provide a personalized career roadmap
   Generate a step-by-step improvement plan including skill development, project suggestions, and profile enhancement strategies based on user goals.

6. To build a user-friendly web-based interface
   Develop an accessible and intuitive platform where users can upload documents, view reports, and track improvements efficiently.

7. To improve employability outcomes
   Help users systematically enhance their resumes and increase their chances of passing ATS screening and securing job opportunities.

Chapter-4

SYSTEM ARCHITECTURE

SkillVerse AI follows a **modular, web-based client-server architecture** integrated with AI services for intelligent resume analysis and recommendation generation.

---

1. User Interface Layer (Frontend)

- Web-based interface built using technologies like HTML, CSS, JavaScript (or React).
- Allows users to:
  - Upload resumes (PDF/DOCX)
  - Upload job descriptions
  - View ATS score, reports, and recommendations
- Ensures smooth user interaction and visualization of results.

---

2. Application Layer (Backend Server)

- Handles all core logic and request processing.
- Built using frameworks like Node.js / Flask / Django.
- Responsibilities:
  - File handling (upload, parsing)
  - Request validation
  - Communication with AI APIs
  - Generating structured reports

---

3. Resume & Job Description Processing Module

- Extracts text from uploaded files using parsing libraries.
- Cleans and preprocesses data (removing noise, formatting text).
- Converts unstructured content into analyzable format.

---

4. AI Analysis Module (Gemini API Integration)

- Core intelligence of the system.
- Uses Gemini API to:
  - Analyze resume content
  - Evaluate ATS compatibility
  - Compare resume with job description
- Generates:
  - ATS score
  - Keyword match analysis
  - Content improvement suggestions

---

5. Recommendation Engine

- Processes AI output and structures it into:
  - Resume improvement suggestions
  - Missing skills identification

    - Keyword optimization tips
- Generates personalized roadmap:
    - Skills to learn
    - Projects to build
    - Profile enhancement steps

---

6. Database Layer

- Stores:
    - User data (optional login system)
    - Uploaded resumes (or metadata)
    - Analysis reports and history
- Can use MongoDB / MySQL depending on design.

---

7. Output & Visualization Module

- Displays results in structured format:
    - ATS score (numeric + visual)
    - Section-wise feedback
    - Job-match analysis
    - Roadmap timeline
- Improves user understanding through clear UI representation.

---

8. System Flow (Data Flow Summary)

1. User uploads resume (and optionally job description).
2. Backend receives and processes files.
3. Text is extracted and cleaned.
4. Data is sent to Gemini API for analysis.
5. AI returns evaluation and suggestions.
6. Recommendation engine refines output.
7. Results are stored and displayed to the user.

Chapter-5

TECHNOLOGIES USED

1. Frontend Technologies

- **HTML, CSS, JavaScript**
  Used to design the user interface for uploading resumes, viewing reports, and interacting with the system.
- **React.js (optional but recommended)**
  Enables dynamic UI, faster rendering, and better user experience for displaying analysis results.

---

2. Backend Technologies

- **Node.js / Express.js (or Flask/Django)**
  Handles server-side logic, API requests, file processing, and communication with the AI module.
  → *Pick one. Don't write all three in your report.*

---

3. AI Integration

- **Gemini API**
  Core component used for:
  - Resume analysis
  - ATS scoring
  - Job description comparison
  - Generating recommendations and roadmap

---

4. File Processing

- **PDF/DOCX Parsing Libraries** (e.g., pdf-parse, docx, PyMuPDF)
  Extracts text from uploaded resumes and job descriptions for analysis.

---

5. Database

- **MongoDB (NoSQL)**
  Stores user data, uploaded files (or references), and generated reports.
  (Alternative: MySQL if structured storage is required.)

---

6. Natural Language Processing (NLP)

- **Basic NLP Techniques / Libraries** (if used alongside Gemini)
  - Keyword extraction
  - Text preprocessing

---

7. Hosting & Deployment

- **Cloud Platforms (e.g., Vercel / Netlify / AWS / Render)**

  Used to deploy frontend and backend for real-time access.

---

8. Version Control

- **Git & GitHub**

  Used for source code management, collaboration, and version tracking.

Chapter-6

METHODOLOGY

The development of SkillVerse AI follows a systematic approach to process user inputs, perform AI-based analysis, and generate actionable outputs. The methodology is divided into the following stages:

---

1. Data Input Stage

- The user uploads:
  - Resume (PDF/DOCX)
  - Optional job description
- Input validation is performed to ensure correct file format and size.

---

2. Data Extraction & Preprocessing

- Text is extracted from uploaded documents using parsing libraries.
- Preprocessing steps include:
  - Removing special characters and noise
  - Normalizing text
  - Structuring sections (skills, education, experience)

---

3. Resume Analysis (AI Processing)

- The processed resume data is sent to the Gemini API.
- The AI performs:
  - ATS compatibility evaluation
  - Keyword analysis
  - Content quality assessment
- An ATS score is generated along with detailed feedback.

---

4. Job Description Matching (Optional Module)

- If a job description is provided:
  - The system compares resume content with job requirements
  - Identifies:
    - Missing skills
    - Keyword gaps
    - Role misalignment

---

5. Recommendation Generation

- Based on AI output, the system generates:
  - Resume improvement suggestions
  - Keyword optimization tips
  - Section-wise corrections
- Suggestions are specific and actionable.

6. Roadmap Creation

- The system builds a personalized roadmap including:
  - Skills to learn
  - Relevant projects to work on
  - Profile enhancement strategies
- Tailored according to user goals and analysis results.

7. Result Presentation

- The final output is displayed to the user in a structured format:
  - ATS score (numeric + visual)
  - Detailed report
  - Job-match analysis
  - Improvement roadmap

8. Data Storage (Optional)

- User data and reports are stored in a database for:
  - Future access
  - Progress tracking

CHAPTER 7

IMPLEMENTATION

7.1 Introduction

The implementation of SkillVerse AI is carried out using a modular full-stack approach, integrating frontend, backend, and AI services to achieve seamless resume analysis and recommendation generation.

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1. Frontend Implementation

- Developed using **React.js (or HTML, CSS, JavaScript)**.
- Key features implemented:
  - Resume upload interface (PDF/DOCX)
  - Job description input field/upload
  - Dashboard to display ATS score and reports
- Used component-based design for:
  - File upload module
  - Report visualization (charts, sections)
  - Roadmap display

---

2. Backend Implementation

- Built using **Node.js with Express.js** (or Flask/Django if Python-based).
- Core functionalities:
  - API endpoints for file upload and analysis
  - Request handling and validation
  - Integration with Gemini API
- Example endpoints:
  - /upload-resume
  - /analyze
  - /compare-job

---

3. File Processing Implementation

- Integrated libraries such as:
  - **pdf-parse / PyMuPDF** for PDF extraction
  - **docx parser** for Word documents
- Extracted raw text is:

- Cleaned
- Structured into sections (skills, experience, education)

---

4. AI Integration (Gemini API)

- Resume data is sent to Gemini API using structured prompts.
- Prompts are designed to:
  - Generate ATS score
  - Identify missing keywords

---

5. Recommendation Engine Implementation

- Processes AI output to:
  - Categorize feedback (skills, formatting, content)
  - Generate actionable suggestions
- Implements logic to:
  - Highlight missing skills from job description
  - Suggest keyword additions

---

6. Roadmap Generation Module

- Based on analysis results:
  - Maps missing skills to learning resources
  - Suggests projects relevant to user goals
- Outputs a step-by-step improvement plan.

---

7. Database Implementation

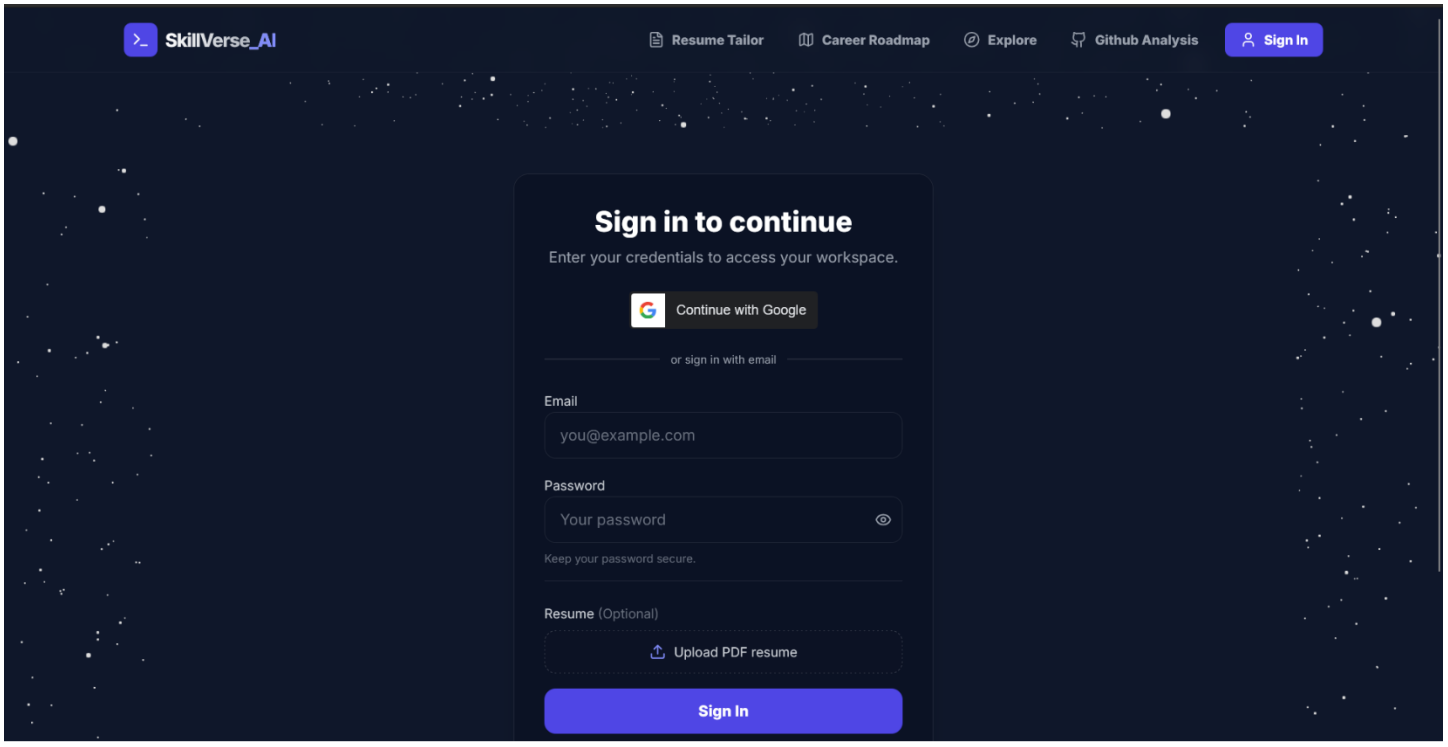
- Used **MongoDB** to store:
  - User details (if login system exists)
  - Uploaded file references
  - Generated reports
- Enables retrieval of past analysis and tracking improvements.

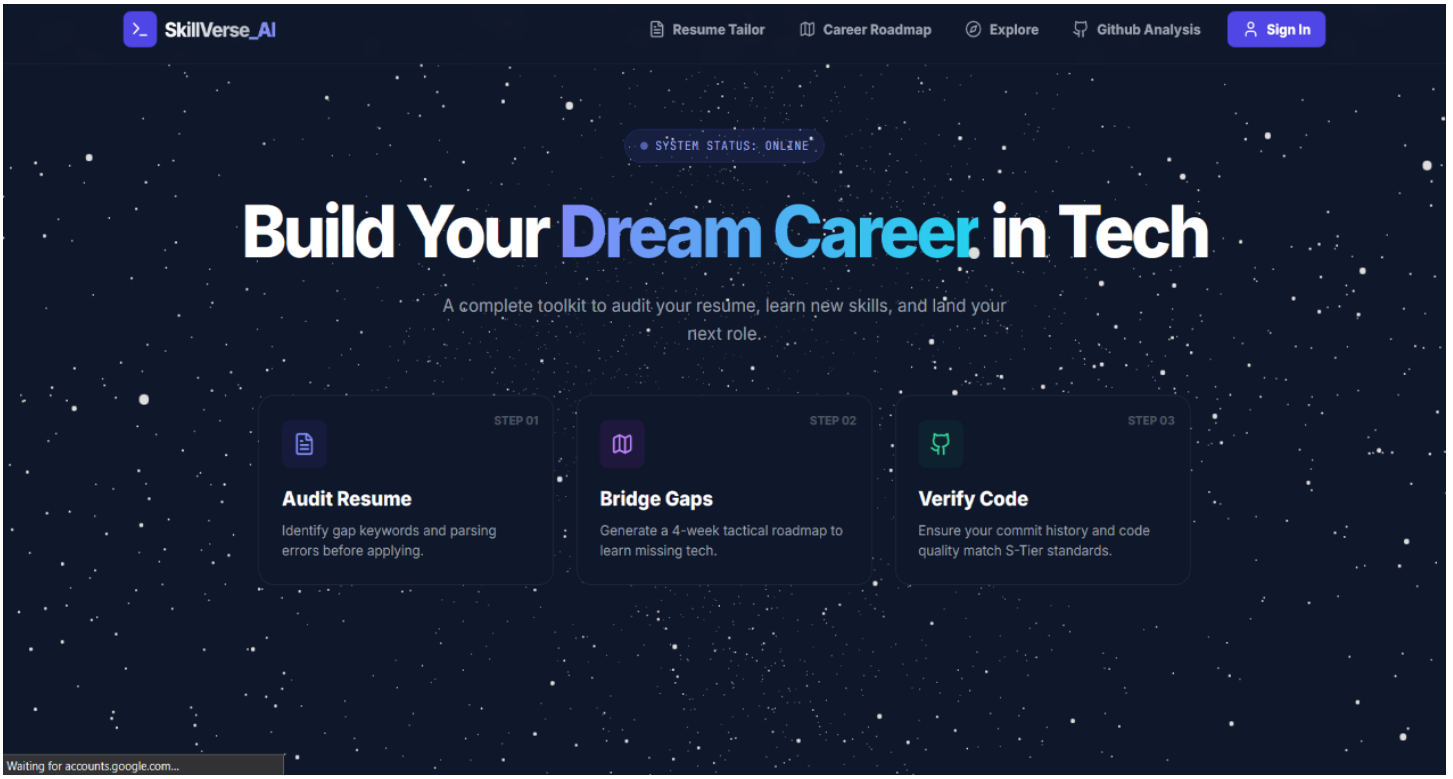
---

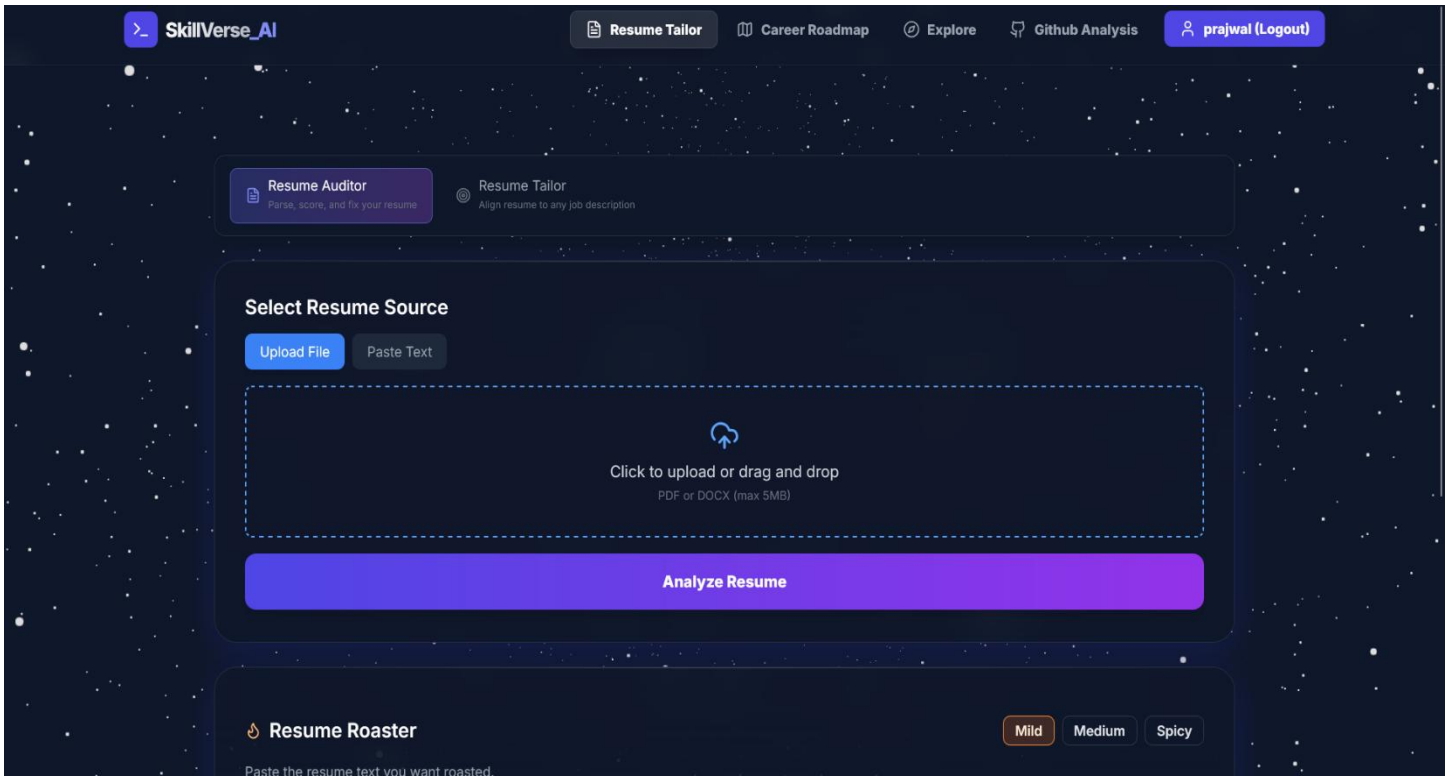
8. Integration Workflow

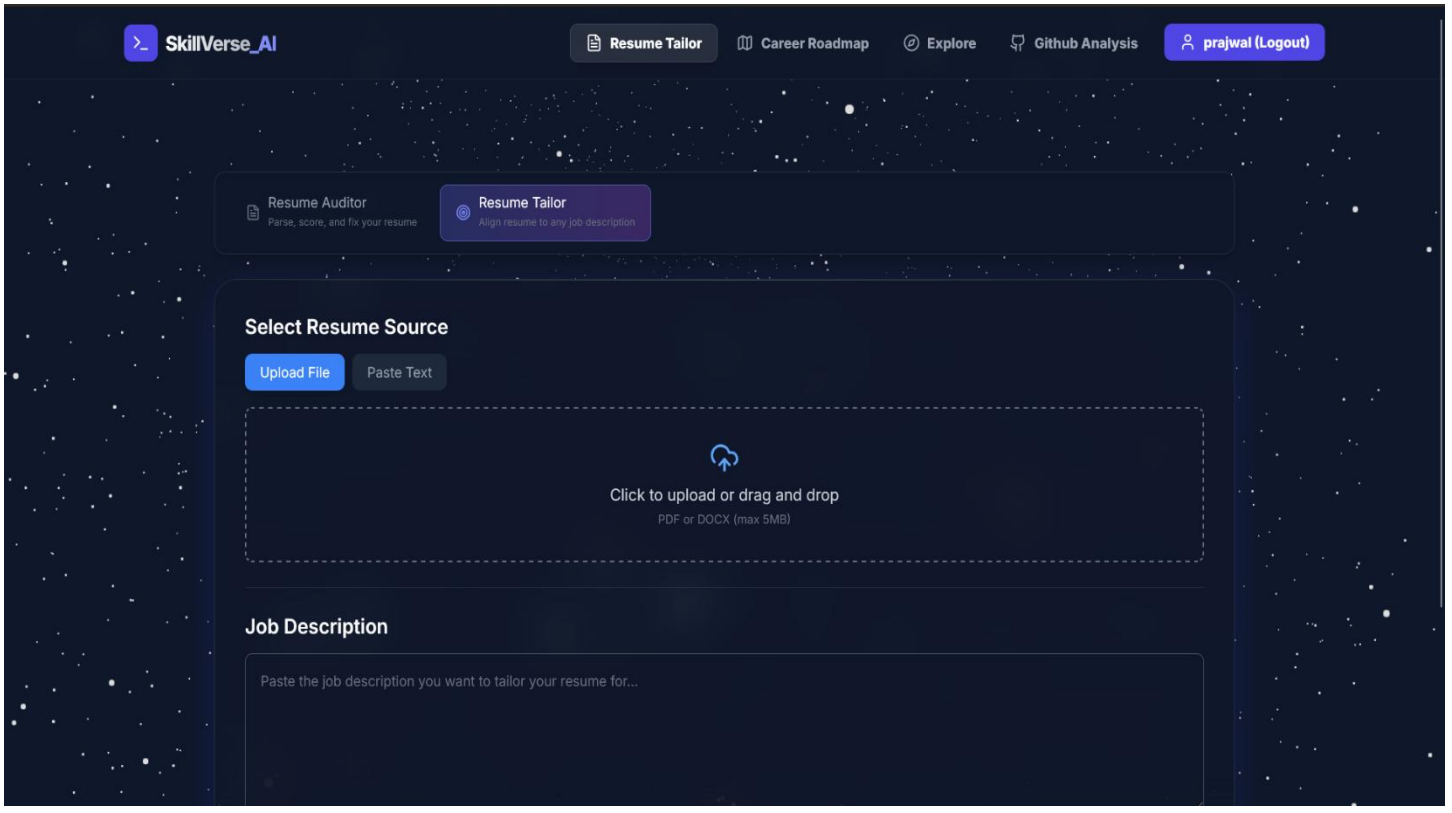
1. User uploads resume/job description via frontend.
2. Backend receives and processes files.
3. Text is extracted and preprocessed.
4. Data is sent to Gemini API.
5. AI response is processed and structured.
6. Recommendations and roadmap are generated.
7. Results are stored and displayed to the user.

RESULT Chapter-8






SkillVerse_AI
Resume Tailor
Career Roadmap
Explore
Github Analysis
prajwal (Logout)
Resume Auditor
Parse, score, and fix your resume
Resume Tailor
Align resume to any job description
Select Resume Source
Upload File
Paste Text
Click to upload or drag and drop
PDF or DOCX (max 5MB)
Analyze Resume
Resume Roaster
Mild
Medium
Spicy
Paste the resume text you want roasted.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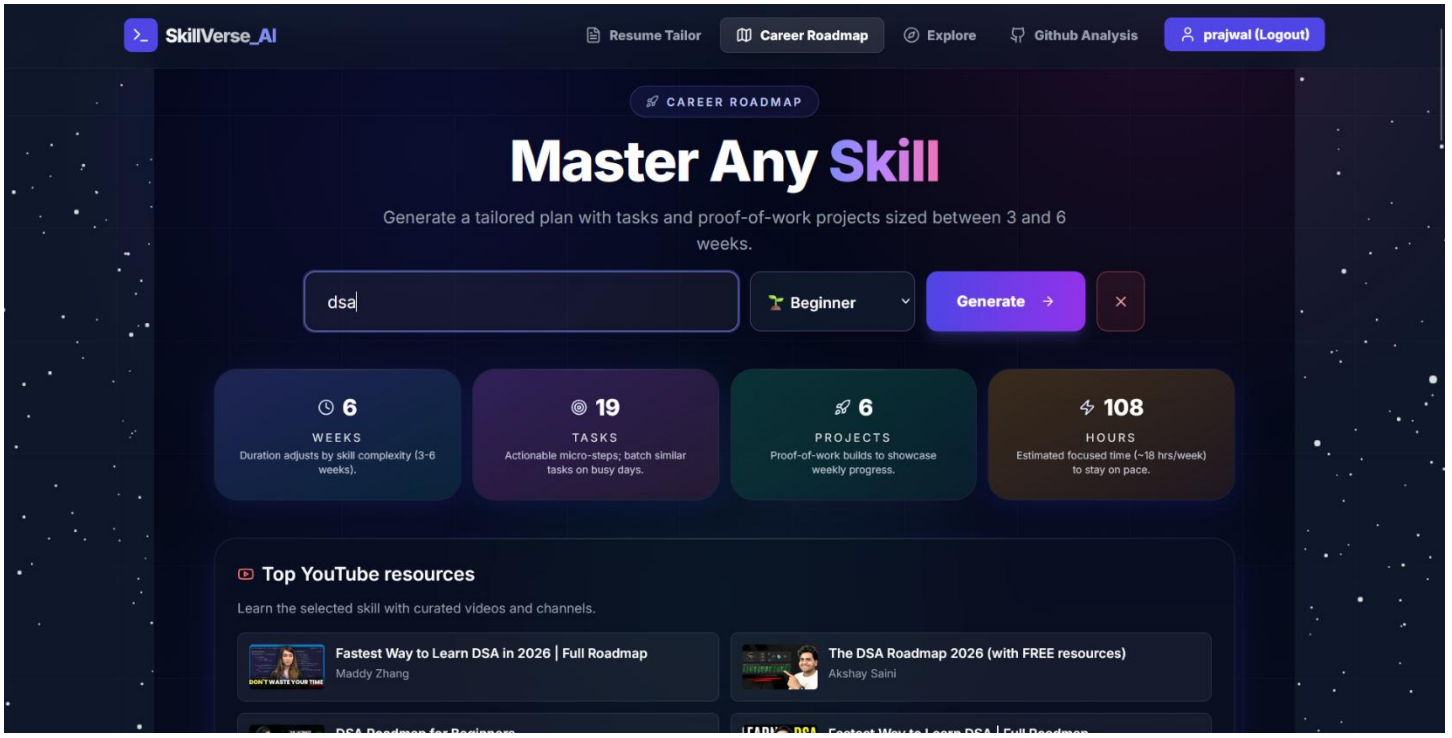
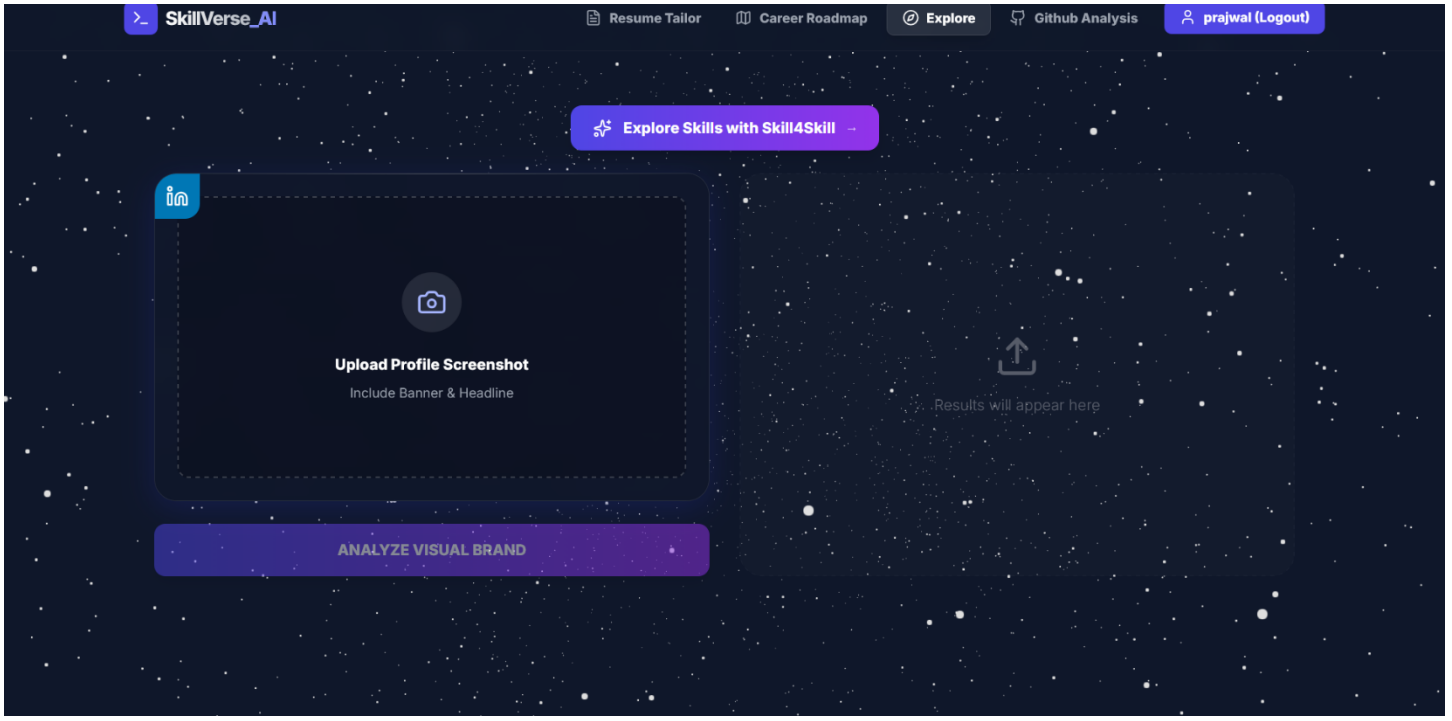
Select Resume Source
Upload File
Paste Text
Click to upload or drag and drop
PDF or DOCX (max 5MB)
Job Description
Paste the job description you want to tailor your resume for...


SkillVerse_AI
Resume Tailor
Career Roadmap
Explore
Github Analysis
prajwal (Logout)
CAREER ROADMAP
Master Any Skill
Generate a tailored plan with tasks and proof-of-work projects sized between 3 and 6 weeks.
dsa
Beginner
Generate
6
WEEKS
Duration adjusts by skill complexity (3-6 weeks).
19
TASKS
Actionable micro-steps; batch similar tasks on busy days.
6
PROJECTS
Proof-of-work builds to showcase weekly progress.
108
HOURS
Estimated focused time (~18 hrs/week) to stay on pace.
Top YouTube resources
Learn the selected skill with curated videos and channels.
Fastest Way to Learn DSA in 2026 | Full Roadmap
Maddy Zhang
The DSA Roadmap 2026 (with FREE resources)
Akshay Saini


SkillVerse_AI
Resume Tailor
Career Roadmap
Explore
Github Analysis
prajwal (Logout)
Explore Skills with Skill4Skill
Upload Profile Screenshot
Include Banner & Headline
Results will appear here
ANALYZE VISUAL BRAND

Chapter-9

ADVANTAGES & LIMITATIONS

Advantages:

1. Fast AI-Based Resume Analysis
   Provides instant evaluation and feedback, reducing manual effort.
2. ATS-Focused Optimization
   Improves chances of passing automated resume screening systems.
3. Job Description Matching
   Identifies gaps between resume and job requirements with precise suggestions.
4. Actionable Recommendations
   Gives clear, practical improvements instead of vague feedback.
5. Personalized Career Roadmap
   Guides users on skills, projects, and profile enhancement.

Limitations:

1. AI Output May Be Inconsistent
   Results depend on Gemini API quality and prompt design.
2. Not Fully Accurate for All ATS Systems
   Different companies use different ATS algorithms.
3. Parsing Issues with Complex Resumes
   Poor formatting can lead to incorrect analysis.
4. Limited Context Understanding
   AI may misinterpret domain-specific experience.
5. Data Privacy Risks
   Handling sensitive resume data requires strong security measures.

Chapter-10

FUTURE ENHANCEMENT

1. **Advanced ATS Simulation Engine**
   Develop a more accurate ATS scoring model by incorporating multiple real-world ATS patterns and rule-based evaluation.

2. **Real-Time Job Market Integration**
   Integrate job APIs (LinkedIn, Indeed) to provide up-to-date skill recommendations based on current market demand.

3. **Resume Auto-Generation & Editing**
   Enable users to automatically generate or rewrite resumes using AI with optimized keywords and structure.

4. **Portfolio & GitHub Analysis**
   Extend analysis beyond resumes by evaluating GitHub profiles and portfolios to provide deeper insights into technical skills.

5. **Multi-Language Support**
   Allow resume analysis in multiple languages to support a wider range of users globally.

Chapter-11

CONCLUSION

SkillVerse AI successfully addresses the challenges faced by job seekers in optimizing their resumes for modern recruitment systems. By leveraging the Gemini API, the system provides automated resume analysis, ATS compatibility scoring, and job-specific recommendations, enabling users to identify and correct gaps in their profiles. The integration of job description matching and personalized roadmap generation transforms the platform from a simple evaluation tool into a comprehensive career enhancement system.

The project demonstrates how artificial intelligence can be effectively applied to solve real-world problems in recruitment by delivering structured insights and actionable guidance. It reduces dependency on manual review, improves resume quality, and increases the likelihood of candidates passing initial screening processes.

Overall, SkillVerse AI serves as a scalable and practical solution that bridges the gap between candidate profiles and industry expectations, with strong potential for future expansion into a complete career development ecosystem.

Chapter-12

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- **MongoDB Documentation. Available: https://www.mongodb.com/docs/**
- **GitHub Guides. Available: https://github.com/pdprajwal9522-cloud/ibm_project.git**
- **, Research Articles and Online Resources.**